

WPP Media Projects

Note: This document keeps the original structure and formatting style, updated to reflect the latest, interview-safe context and ownership boundaries.

Project 1: Audience Profiling, Audience Insights, and Audience Creation Platform

Tech Stack

- Machine Learning: downstream propensity scoring workflows, model-based attribute importance (where applicable)
- GenAI: LLM-assisted narrative summaries for charts and tables
- Tools: Python, Dash, Snowflake, SQL, Git, CI/CD; retrieval and context grounding integrated via Snowflake information schema plus feature dictionaries (owned by other teammates)

Project Description

At WPP Media, I helped build and evolve a production internal Audience Creation and Audience Profiling platform used by analytics teams to define, validate, and reuse audiences, then profile them through repeatable insights. The platform replaced manual, inconsistent SQL workflows with a governed workflow that generates validated Snowflake SQL and supports large-scale audience exploration.

Responsibilities

- Owned the end-to-end Audience Creation module, including the Dash UI, rule builder, and the deterministic SQL compiler supporting nested AND/OR rule trees
- Designed and optimized Snowflake SQL generation patterns to handle high-cardinality filtering, joins, aggregations, and feature engineering in a safe, repeatable way
- Implemented validation and guardrails to enforce consistent joins, time windows, allowed operators, and policy-aware feature usage, preventing invalid or unsafe queries
- Introduced LLM-assisted interpretability for profiling outputs by creating prompt templates per visualization type to generate concise, grounded summaries on top of deterministic charts and tables
- Worked with analysts to collect feedback, validate correctness versus trusted manual SQL, and iterate on usability and edge cases

Result

- Deployed and tested with an internal analytics team of about 15 users, supporting real workflows and iterative feedback cycles
- Reduced time-to-first-audience and profiling iterations, often from 30 to 60+ minutes manually to under about 10 minutes using the governed workflow, based on repeated internal comparisons

- Improved consistency of business outcomes by standardizing joins, time windows, and validation logic so similar questions produced comparable audiences
- Improved accessibility by enabling less SQL-heavy users to define sophisticated audiences through a guided UI and validated SQL generation
- Established a foundation for scalable ML and GenAI extensions while preserving governance and trust

Project 2: Audience Propensity and Media Response Modeling for Global Brands

Tech Stack

- Machine Learning Models: propensity models; media response analysis, including time series approaches when applicable
- AutoML Platform: DataRobot
- Tools: Python, Snowflake, SQL, feature engineering pipelines, cloud tooling

Project Description

At WPP Media, I supported the development and operationalization of large-scale audience propensity and related modeling workflows for major global brands. The work enabled more data-driven audience selection and activation by producing consistent scoring outputs and repeatable feature engineering pipelines.

Responsibilities

- Performed feature engineering and feature selection across large-scale consumer datasets, using transactional, demographic, and behavioral signals stored in Snowflake
- Used DataRobot AutoML to train and evaluate multiple model blueprints, iterating based on diagnostics such as feature importance, partial dependence, and error analysis
- Partnered with analytics and activation stakeholders to validate model outputs against business objectives and ensure scores were interpretable and stable
- Produced batch scoring outputs and documentation to support downstream activation and measurement workflows

Post-deployment measurement and feedback loop (Downstream Measurement Stack, NOT performed by me)

After deploying and scoring, outputs are consumed by downstream measurement systems to support activation and planning, with a feedback loop from observed results back into the measurement stack. I am NOT involved in this process.

- MMM: Analyze results, feed back to recalibrate MMM, and generate updated budget recommendations for activation and planning. As actualized data becomes available,

incremental lift and performance are estimated and surfaced in Analyze Results or a dedicated MMM dashboard tab.

- PM: Follows the same approach as MMM, using analyzed results and actualized data to recalibrate and refresh recommendations for planning and optimization.
- MTM: Analyze results feed into MTM to re-rank media metrics across touchpoints, sub-touchpoints, and segments for in-platform optimization during activation. As actualized data becomes available, lift is estimated via the opportunity index within the MTM dashboard.

Measurement framework flow (high level)

Downstream Measurement Stack (PM, MTM, MMM) → Activation & Planning

Analyze Results ↻ feeds back into Measurement Stack (recalibration and re-ranking)

Downstream Measurement Stack → Display Results (dashboard surfacing)

Result

- Enabled more consistent audience scoring workflows to support targeting and activation decisions
- Delivered repeatable modeling pipelines that adapted to varied client and category use cases

Project 3: Fusion-as-a-Service (FaaS) for Survey-to-Population Enrichment

Tech Stack

- Algorithms: hot-deck style nearest-neighbor fusion (kNN), donor-recipient matching
- Techniques: distance-based matching, probabilistic donor selection, dimensionality reduction (PCA) where applicable, automated hyperparameter tuning (Autofusion), genetic optimization for predictor weighting, NLP-assisted feature matching for hook mapping
- Tools: Python, Snowflake, Google Cloud Storage, Kubeflow, SQL, feature engineering pipelines

Project Description

At WPP Media, I worked on delivering Fusion-as-a-Service (FaaS), a production data enrichment pipeline designed to project survey-level insights onto a larger population dataset. Fusion operates like an ML-style similarity and optimization pipeline: it uses kNN / hot-deck matching to identify similar donor records using shared “hook” variables, then transfers survey target attributes to matched recipient records through statistically grounded, probabilistic selection. To improve match quality and stability, the system supports automated tuning of hot-deck parameters (Autofusion) and optimization of predictor weights, including genetic-algorithm based search. ML-adjacent components, such as PCA (to manage high-dimensional feature spaces) and NLP-assisted standardization of string variables (to suggest or validate hook mappings across datasets), help make the enrichment workflow scalable and repeatable across markets and survey instruments.

Responsibilities

- Prepared and aligned donor and recipient datasets, supporting schema and hook mapping, feature engineering, and standardized data dictionaries
- Executed and iterated on the fusion pipeline using kNN / hot-deck style matching, validating quality with distributional alignment checks, stability across slices, and sensitivity to key configuration settings
- Supported automated tuning workflows (Autofusion) across parameters such as k (neighbors), match limits, distance metrics, and distance-weighted selection, plus optimization of predictor weights to improve matching quality
- Applied PCA where applicable to reduce dimensionality and improve matching behavior in high-feature settings; supported NLP-assisted mapping and validation for string-based variables used in hook alignment
- Supported pipeline execution and orchestration in a GCP-native environment using Kubeflow and cloud storage, and produced production-ready outputs plus data maps to preserve interpretability of enriched variables

Result

- Enabled population-scale inference from survey samples by enriching recipient populations with survey-derived attributes via statistically grounded similarity matching and automated optimization
- Delivered a reusable, scalable enrichment workflow applicable across markets and survey instruments, with interpretable data maps and repeatable validation patterns

Project 4: Experiment and Measurement App (EMA) for Geo-Based Media Incrementality Testing

Tech Stack

- GenAI and Agents: Vertex AI agents for conversational experiment design and measurement, prompt and context engineering, agent-driven translation of plain-language intent into structured experiment parameters
- Experiment Design: GeoFlex (Google) for automated experiment design exploration (market matching, MDE computation, power analysis, design optimization), integrated alongside custom in-house experimentation methods
- Tools: React, TypeScript, FastAPI, Python, Vertex AI, GCP, pytest

Project Description

At WPP Media, I helped design and build a production platform that automates the end-to-end workflow of geo-based media incrementality experiment design and measurement. Traditionally, designing a geo experiment requires deep statistical expertise for selecting test and control markets, computing minimum detectable effects, balancing budget and duration constraints, and validating designs for contamination and power. EMA automates all of this using Google's GeoFlex library and WPP's custom experimentation methods, and delivers it through an agentic experience that guides non-technical marketing leaders through every step. The platform uses a split-screen interface where a conversational agent collects the user's business context, triggers automated design exploration, and delivers results in business terms on one side, while their responses map to editable, validated fields on the other. EMA supports two workflows, one for designing and configuring an experiment and one for evaluating results after the experiment runs.

Responsibilities

- Helped design the overall agentic system and built parts of the end-to-end agentic workflow across both the exploration agent for experiment design and the measurement agent for post-experiment evaluation
- Built the QA layer that validates user-provided data for consistency, completeness, and alignment with uploaded datasets, and surfaces contradictions and tradeoffs back to the user during the conversation
- Developed both frontend and backend for the modules I owned, using React and TypeScript on the client and FastAPI on the server, mapping conversational responses to structured questionnaire objects in real time

- Integrated GeoFlex along with custom in-house methods to power automated design exploration, including market matching, MDE computation, and budget/duration optimization to surface viable experiment designs
- Set up a shared context layer so the team could use Claude efficiently and consistently throughout development
- Wrote pytest coverage for the features I built to keep the workflows reliable as the platform evolved

Result

- Replaced an earlier form-first wizard flow and Streamlit prototype with a conversational, agent-driven experience designed for non-technical marketing leaders
- Automated statistical workflows that previously required deep experimentation expertise, enabling client-side users to configure valid geo experiments through plain-language interaction
- Delivered a unified split-screen paradigm across both the design and measurement workflows, keeping the agent interaction and the structured questionnaire in sync
- Established reusable QA, validation, and context patterns that supported consistent experiment setup across varied client goals and verticals